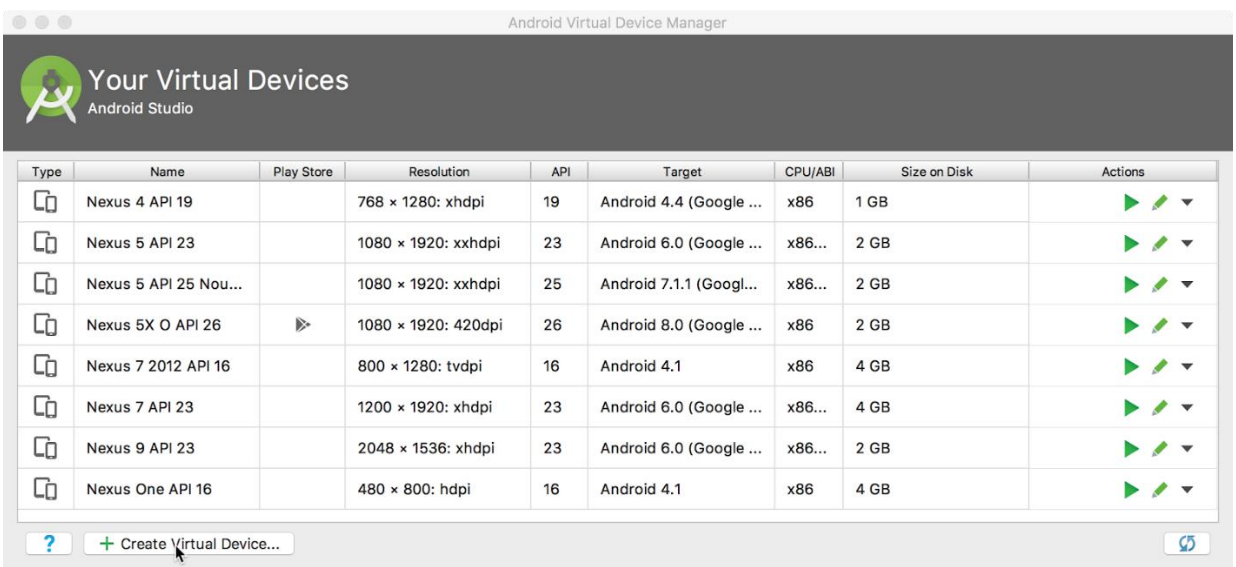
Use a virtual device (emulator)

In this task, you will use the Android Virtual Device (AVD) manager to create a virtual device (also known as an emulator) that simulates the configuration for a particular type of Android device, and use that virtual device to run the app. Note that the Android Emulator has additional requirements beyond the basic system requirements for Android Studio. Using the AVD Manager, you define the hardware characteristics of a device, its API level, storage, skin and other properties and save it as a virtual device. With virtual devices, you can test apps on different device configurations (such as tablets and phones) with different API levels, without having to use physical devices.

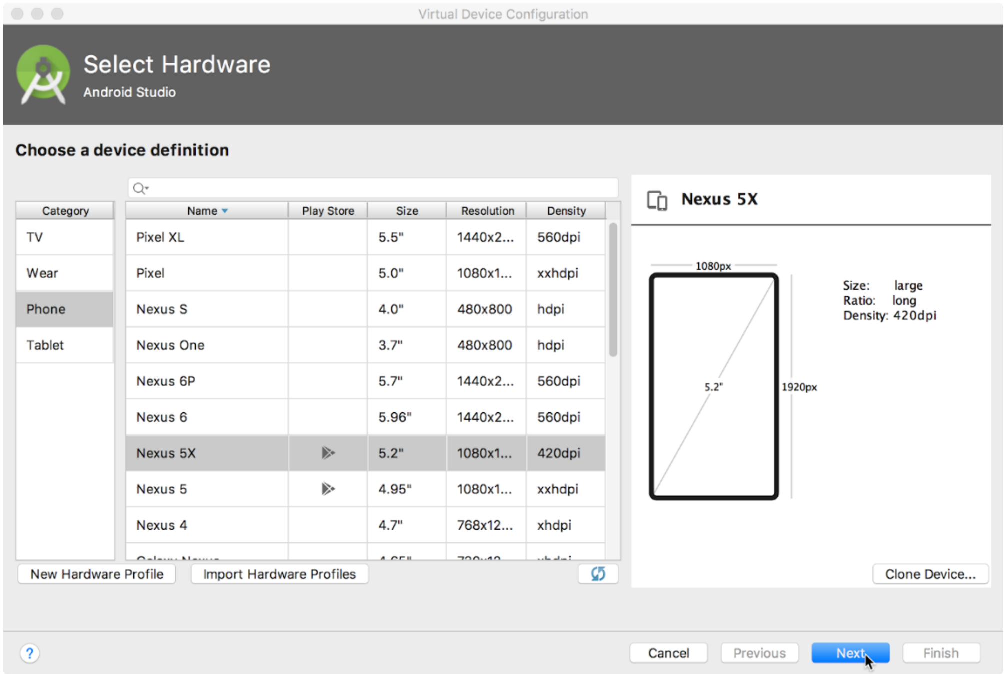
1 Create an Android virtual device (AVD)

In order to run an emulator on your computer, you have to create a configuration that describes the virtual device.

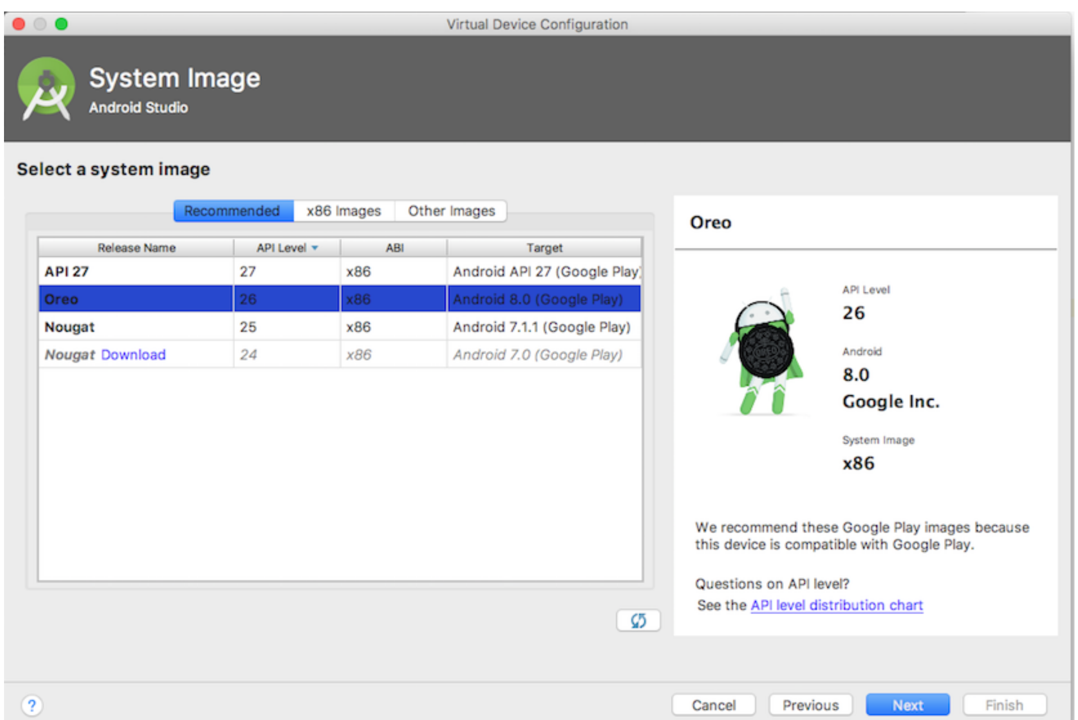
1. In Android Studio, select **Tools > Android > AVD Manager**, or click the AVD Manager icon  in the toolbar. The **Your Virtual Devices** screen appears. If you've already created virtual devices, the screen shows them (as shown in the figure below); otherwise you see a blank list.



2. Click the **+Create Virtual Device**. The **Select Hardware** window appears showing a list of pre configured hardware devices. For each device, the table provides a column for its diagonal display size (**Size**), screen resolution in pixels (**Resolution**), and pixel density (**Density**).



3. Choose a device such as **Nexus 5x** or **Pixel XL**, and click **Next**. The **System Image** screen appears.
4. Click the **Recommended** tab if it is not already selected, and choose which version of the Android system to run on the virtual device (such as **Oreo**).



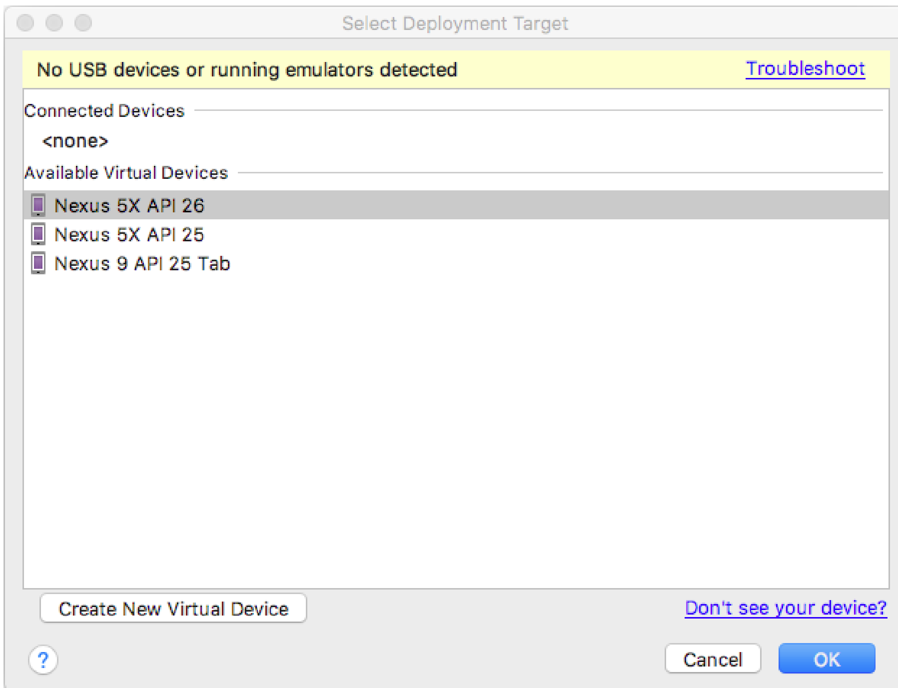
There are many more versions available than shown in the **Recommended** tab. Look at the **x86 Images** and **Other Images** tabs to see them. If a **Download** link is visible next to a system image you want to use, it is not installed yet. Click the link to start the download, and click **Finish** when it's done.

5. After choosing a system image, click **Next**. The **Android Virtual Device (AVD)** window appears. You can also change the name of the AVD. Check your configuration and click **Finish**.

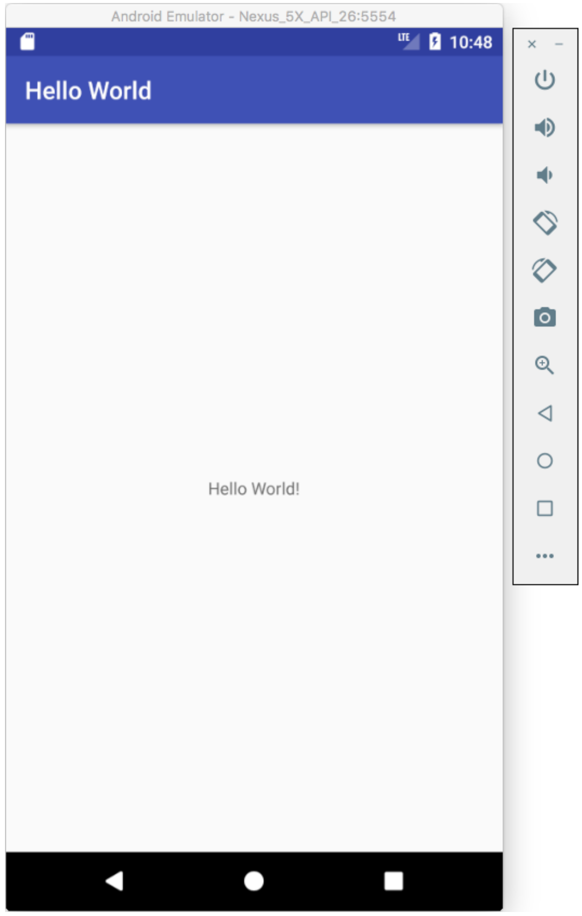
2 Run the app on the virtual device

In this task, you will finally run your Hello World app.

1. In Android Studio, choose **Run > Run app** or click the **Run** icon  in the toolbar.
2. The **Select Deployment Target** window, under **Available Virtual Devices**, select the virtual device, which you just created, and click **OK**.



The emulator starts and boots just like a physical device. Depending on the speed of your computer, this may take a while. Your app builds, and once the emulator is ready, Android Studio will upload the app to the emulator and run it. You should see the Hello World app as shown in the following figure.


Android Emulator - Nexus_5X_API_26:5554
10:48
Hello World
Hello World!